

# Technical Document #207: Cloud Computing - Comprehensive Overview

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Content delivery networks (CDNs) distribute content globally to reduce latency. CloudFront and Cloudflare are popular CDN providers.

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

Multi-cloud strategies use services from multiple cloud providers. They reduce vendor lock-in and optimize for cost and performance.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

### Key Takeaways:

- Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files.
- Content delivery networks (CDNs) distribute content globally to reduce latency.
- Auto-scaling automatically adjusts compute resources based on demand.